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3 Complex Analysis 2023

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Final Exam

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Instructions:

- This question paper consists of **8 pages** (including this one).
- You have 120 minutes to answer the questions.
- Answer all questions in the spaces provided on this question paper. Use the backs of pages if needed.
- Use your *UCT Examination Answer Book* for rough work. The work in your UCT Examination Answer Book will not be marked. Only the answers that you write on this question paper will be marked.
- Give full proofs.
- Be careful to provide answers that we can read and make sense of at all times. If we can't read your handwriting, we will mark your solution incorrect.
- 60 points = 100% (63 points available).

Problem	1	2	3	4	5	6	Σ	Mark
Maximum Points	12	8	10	13	10	10	60	100%
Result								

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Problem 1

(12 points)

- (a) Find the set of all $z \in \mathbb{C}$ where the function $f(z) = z \operatorname{Re} z$ is differentiable.
- (b) Find the radius of convergence of the power series $\sum_{n=1}^{\infty} \frac{z^{2n}}{\sqrt{n}}$.

Solution.

- (a) We first show that f is not differentiable at any $z \neq 0$.

Let $z = x + iy$ with $x = \operatorname{Re} z$, $y = \operatorname{Im} z$, and let $f = u + iv$ with $u = \operatorname{Re} f$, $v = \operatorname{Im} f$. We have

$$f(z) = z \operatorname{Re} z = (x + iy)x = x^2 + ixy,$$

and therefore

$$u(x, y) = x^2, \quad v(x, y) = xy.$$

We have

$$u_x(x, y) = 2x, \quad u_y(x, y) = 0, \quad v_x(x, y) = y, \quad v_y(x, y) = x.$$

The Cauchy-Riemann equations $u_x = v_y$ and $u_y = -v_x$ take the form

$$2x = x, \quad 0 = -y.$$

They can only be fulfilled if $x = 0$ and $y = 0$. Thus, f is not differentiable at any $z \neq 0$.

Now let us consider $z = 0$. We have

$$\lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{h \operatorname{Re} h - 0}{h} = \lim_{h \rightarrow 0} \operatorname{Re} h = 0.$$

Thus, f is differentiable at 0 (with $f'(0) = 0$).

- (b) *1st solution.* We are going to use the Ratio Test for the real series $\sum_{n=1}^{\infty} \frac{|z|^{2n}}{\sqrt{n}}$. We have

$$L = \lim_{n \rightarrow \infty} \frac{|z|^{2n+2}}{\sqrt{n+1}} \frac{\sqrt{n}}{|z|^{2n}} = |z|^2 \lim_{n \rightarrow \infty} \frac{\sqrt{n}}{\sqrt{n+1}} = |z|^2 < 1 \iff |z| < 1.$$

The radius of convergence of the series $\sum_{n=1}^{\infty} \frac{z^{2n}}{\sqrt{n}}$ is $R = 1$.

2nd solution. The coefficients of the power series $\sum_{n=1}^{\infty} \frac{z^{2n}}{\sqrt{n}}$ are

$$a_m = \begin{cases} \frac{1}{\sqrt{\frac{m}{2}}} = \sqrt{\frac{2}{m}}, & m \text{ is even,} \\ 0, & m \text{ is odd.} \end{cases}$$

By the Cauchy-Hadamard formula,

$$\frac{1}{R} = \limsup_{m \rightarrow \infty} |a_m|^{\frac{1}{m}} = \lim_{m \rightarrow \infty} \left(\sqrt{\frac{2}{m}} \right)^{\frac{1}{m}} = \lim_{m \rightarrow \infty} \left(\frac{2}{m} \right)^{\frac{1}{2m}} = \lim_{m \rightarrow \infty} e^{\frac{1}{2m} \ln \left(\frac{2}{m} \right)} = e^0 = 1.$$

Thus, $R = 1$.

Problem 2

(8 points)

Calculate the integral $\int_{\gamma_k} \frac{e^{z^2+1}}{(z-i)^2} dz$, $k = 1, 2$, where

(a) $\gamma_1(t) = 2 + e^{it}$, $t \in [0, 2\pi]$,

(b) $\gamma_2(t) = i + 2e^{it}$, $t \in [0, 2\pi]$.

Solution.

Notice that the function $f(z) = \frac{e^{z^2+1}}{(z-i)^2}$ has the only singularity $z = i$ (which is a pole of order 2).

- (a) The path γ_1 is a circle of radius 1 with center 2, the singularity $z = i$ lies outside of γ_1 . Thus, f is holomorphic in some open disc that contains γ_1 . By the Cauchy Integral Theorem,

$$\int_{\gamma_1} \frac{e^{z^2+1}}{(z-i)^2} dz = 0.$$

- (b) The singularity $z = i$ lies inside the circle γ_2 . We apply the Cauchy Integral Formula for derivatives:

$$\int_{\gamma_2} \frac{e^{z^2+1}}{(z-i)^2} dz = 2\pi i \left(e^{z^2+1} \right)' \Big|_{z=i} = 2\pi i \left(e^{z^2+1} \cdot 2z \right) \Big|_{z=i} = 2\pi i e^{-1+1} 2i = -4\pi.$$

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Problem 3

(10 points)

Assume that the functions f and g are holomorphic in the open disc $D(z_0; R) = \{z \in \mathbb{C} : |z| < R\}$, f has a zero of order $n \in \mathbb{N} \cup \{0\}$ at z_0 , and g has a zero of order $m > n$ at z_0 . Prove that $\frac{f}{g}$ has a pole at z_0 , and determine the order of this pole. Give detailed justifications.

Solution.

By assumptions, the functions f and g can be represented in $D(z_0; R)$ as

$$f(z) = (z - z_0)^n \varphi(z), \quad g(z) = (z - z_0)^m \psi(z),$$

where φ and ψ are holomorphic functions in $D(z_0; R)$ with $\varphi(z_0) \neq 0$, $\psi(z_0) \neq 0$. Then

$$\frac{f(z)}{g(z)} = \frac{1}{(z - z_0)^{m-n}} \frac{\varphi(z)}{\psi(z)}.$$

Since $\psi(z_0) \neq 0$ and ψ is continuous, there exists $r \in (0, R)$ such that $\psi(z) \neq 0$ in $D(z_0; r)$. Then the function $\Phi(z) = \frac{\varphi(z)}{\psi(z)}$ is holomorphic in $D(z_0; r)$, and $\Phi(z_0) \neq 0$. It follows that the function $\frac{f}{g}$ has a pole of order $m - n$ at z_0 .

Problem 4

(13 points)

- (a) Determine the singularities of the function $f(z) = \frac{1}{z^3 - 8}$ in the closed upper half-plane $\{z \in \mathbb{C} : \text{Im } z \geq 0\}$, and calculate the residues at those singularities.
- (b) Give a definition of P.V. $\int_{-\infty}^{\infty} \frac{1}{x^3 - 8} dx$.
- (c) Calculate P.V. $\int_{-\infty}^{\infty} \frac{1}{x^3 - 8} dx$.

Solution.

- (a) The zeros of the polynomial $z^3 - 8$ are third roots of 8, namely,

$$\begin{aligned} z_0 &= 2, \\ z_1 &= 2e^{2\pi i \frac{1}{3}} = 2 \left(\cos \frac{2\pi}{3} + i \sin \frac{2\pi}{3} \right) = -1 + i\sqrt{3}, \\ z_2 &= 2e^{2\pi i \frac{2}{3}} = 2 \left(\cos \frac{4\pi}{3} + i \sin \frac{4\pi}{3} \right) = -1 - i\sqrt{3}. \end{aligned}$$

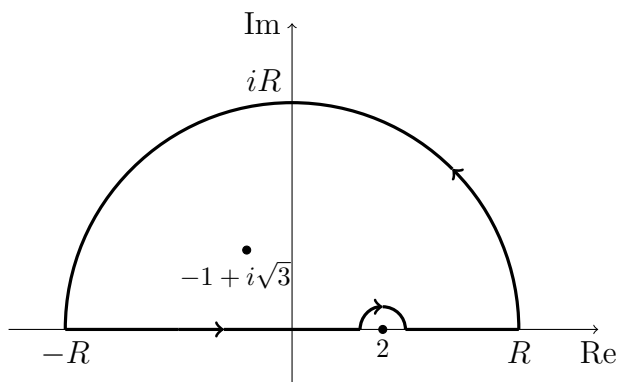
Two of them, namely, $z_0 = 2$ and $z_1 = -1 + i\sqrt{3}$ lie in the closed upper half-plane. They are simple poles. The residues are

$$\text{Res} \left(\frac{1}{z^3 - 8}, 2 \right) = \frac{1}{3z^2} \Big|_{z=2} = \frac{1}{12},$$

$$\begin{aligned} \text{Res} \left(\frac{1}{z^3 - 8}, -1 + i\sqrt{3} \right) &= \frac{1}{3z^2} \Big|_{z=-1+i\sqrt{3}} = \frac{1}{3(-1+i\sqrt{3})^2} = \frac{1}{3(1-2\sqrt{3}-3)} \\ &= -\frac{1}{6(1+i\sqrt{3})} = -\frac{1-i\sqrt{3}}{6(1+3)} = \frac{-1+i\sqrt{3}}{24}. \end{aligned}$$

(b) P.V. $\int_{-\infty}^{\infty} \frac{1}{x^3 - 8} dx = \lim_{\substack{\varepsilon \rightarrow +0 \\ R \rightarrow \infty}} \left(\int_{[-R, 2-\varepsilon]} + \int_{[2+\varepsilon, R]} \right) \frac{1}{x^3 - 8} dx.$

- (c) Consider the path γ indicated below with R large enough:



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Let $\gamma_R(t) = Re^{it}$, $t \in [0, \pi]$ be the large semicircle with radius R , and let $\gamma_\varepsilon(t) = 2 + \varepsilon e^{i(\pi-t)}$, $t \in [0, \pi]$, be a small semicircle with radius ε and center 2, negatively oriented.

We have

$$\int_{\gamma} \frac{1}{z^3 - 8} dz = 2\pi i \operatorname{Res} \left(\frac{1}{z^3 - 8}, -1 + i\sqrt{3} \right) = 2\pi i \frac{-1 + i\sqrt{3}}{24} = \frac{\pi i(-1 + i\sqrt{3})}{12}.$$

On the other hand,

$$\int_{\gamma} \frac{1}{z^3 - 8} dz = \left(\int_{\gamma_R} + \int_{[-R, 2-\varepsilon]} + \int_{\gamma_\varepsilon} + \int_{[2+\varepsilon, R]} \right) \frac{1}{z^3 - 8} dz.$$

If $|z| = R$, then $|z^3 - 8| \geq |z|^3 - 8 = R^3 - 8$. We estimate

$$\left| \int_{\gamma_R} \frac{1}{z^3 - 8} dz \right| \leq \frac{1}{R^3 - 8} \pi R \rightarrow 0, \quad R \rightarrow \infty.$$

Furthermore,

$$\lim_{\varepsilon \rightarrow +0} \int_{\gamma_\varepsilon} \frac{1}{z^3 - 8} dz = -\pi i \operatorname{Res} \left(\frac{1}{z^3 - 8}, 2 \right) = -\frac{\pi i}{12}.$$

It follows that

$$\text{P.V.} \int_{-\infty}^{\infty} \frac{1}{x^3 - 8} dx = \frac{\pi i(-1 + i\sqrt{3})}{12} - \left(-\frac{\pi i}{12} \right) = \frac{\pi i^2 \sqrt{3}}{12} = -\frac{\pi \sqrt{3}}{12}.$$

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Problem 5

(10 points)

Determine the Laurent series of the function $f(z) = \frac{1}{9 + z^2}$ in the given region. Give an explicit formula for the coefficients a_n , $n \in \mathbb{Z}$, of the Laurent series.

(a) $\{z \in \mathbb{C} : |z| < 3\}$,

(b) $\{z \in \mathbb{C} : |z| > 3\}$.

Solution.

(a) We have

$$\frac{1}{9 + z^2} = \frac{1}{9} \cdot \frac{1}{1 - (-\frac{z^2}{9})} = \frac{1}{9} \sum_{n=0}^{\infty} \left(-\frac{z^2}{9}\right)^n = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{9^{n+1}}.$$

The coefficients of the Laurent series are, for $m \in \mathbb{Z}$,

$$a_m = \begin{cases} 0, & m < 0 \text{ or } m \text{ odd,} \\ (-1)^n \frac{1}{9^{n+1}}, & m = 2n \text{ with } n = 0, 1, 2, \dots \end{cases}$$

(b) We have

$$\begin{aligned} \frac{1}{9 + z^2} &= \frac{1}{z^2} \cdot \frac{1}{1 + \frac{9}{z^2}} = \frac{1}{z^2} \cdot \frac{1}{1 - \left(-\frac{9}{z^2}\right)} = \frac{1}{z^2} \sum_{n=0}^{\infty} \left(-\frac{9}{z^2}\right)^n \\ &= \sum_{n=0}^{\infty} (-1)^n 9^n z^{-2n-2} = \sum_{n=1}^{\infty} (-1)^{n-1} 9^{n-1} z^{-2n}. \end{aligned}$$

The coefficients of the Laurent series are, for $m \in \mathbb{Z}$,

$$a_m = \begin{cases} 0, & m \geq 0 \text{ or } m \text{ odd,} \\ (-1)^{n-1} 9^{n-1}, & m = -2n \text{ with } n \in \mathbb{N}. \end{cases}$$

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Problem 6

(10 points)

How many zeros does the polynomial $p(z) = 2z^4 + 2z^3 - 5z + 1$ have in the disc $\{z \in \mathbb{C} : |z| < 2\}$?

Solution.

We consider $f(z) = 2z^4$ and $g(z) = 2z^3 - 5z + 1$. On the circle $|z| = 2$ we have

$$|f(z)| = 2|z|^4 = 32, \quad |g(z)| \leq 2|z|^3 + 5|z| + 1 = 16 + 10 + 1 = 27.$$

Thus,

$$|g(z)| < |f(z)|, \quad |z| = 2.$$

By Rouché's Theorem, f and $p = f + g$ have the same number of zeros in $\{z \in \mathbb{C} : |z| < 2\}$. Clearly, $f(z) = 2z^4$ has 4 zeros (more precisely, a zero at $z = 0$ of order 4) in $\{z \in \mathbb{C} : |z| < 2\}$. We conclude that p has 4 zeros in $\{z \in \mathbb{C} : |z| < 2\}$.